



RESQ-PATENT DRAFT

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RESQ-AN AI ASSISTED DISASTER ALERT VERIFICATION AND RESOURCE ALLOCATION SYSTEM USING A RELATIONAL DATABASE ARCHITECTURE

Field of the idea

This idea sits purely in disaster management and emergency response. It focuses on an intelligent alert system that brings together AI-driven credibility checks, human verification based on role, and a transactional approach to resource allocation, all built on a relational database.

Description of the Problem Statement

When disasters strike i.e. floods, earthquakes, cyclones, wildfires, landslide etc, one need fast, accurate, and coordinated action so that help actually gets to the people who need it. Right now, most disaster alert systems lean on manual reports, crowdsourced tips, or unverified digital info to spot emergencies and hand out resources. But the main issue is that during big disasters(artificial/natural), social media and informal channels explode with rumours, fake alerts, repeat reports, and flat-out false claims. Misinformation spreads fast. This messes everything up and resources get sent to the wrong place, real victims get left waiting, funds disappear, and people lose faith in the whole response system.

Today's present traditional systems usually fall short in the below listed ways:

1. No Credibility Checks: Most alert systems don't use any automated way to check if a disaster alert is real or reliable before escalating it.
2. Too Much Manual Verification: People have to dig through alerts by hand. It's slow, and when the volume spikes, the system can't keep up.
3. No Real Accountability: Lots of platforms skip audit trails and don't keep transactional records when handing out money or supplies, so disputes pop up, data gets messy, and oversight is weak.

4. Poor Role-Based Controls: There's often no clear separation between the people submitting alerts, those verifying them, coordinators, and admins. That opens the door to unauthorized actions.

5. Weak Data Integrity: Systems built on loosely organized data like document stores or client-side storage aren't good for situations where you need strong consistency, traceability, and durability and this really goes well for current resource allocation databases.

All these gaps make it clear—we need a better disaster management system. It has to filter out bad/fake alerts and help people make smart choices, not just leave everything to automation. The backbone should be a tough relational database that keeps data tight, transactions solid, and only lets the right people access sensitive info. When you bring in AI to help with the analysis, but still keep humans in charge, things get clearer. The result is that there is less chaos from misinformation, faster action when a real emergency hits, and relief efforts you can actually track and trust and the records can be easily maintained.

Objectives

The core idea is design and to establish a smart disaster management system that reduces misinformation during natural emergencies while ensuring that relief operations remain swift, accountable, and well-organized.

Its main goals are as follows:

- Ensure that disaster alerts are submitted in a structured format and prevent the widespread dissemination of false information.
- Integrate artificial intelligence to assist in verifying the authenticity of disaster alerts, while keeping crucial decisions under human control.
- Implement a “human-in-the-loop” mechanism so that real individuals always review AI recommendations before any alert is approved or dismissed.
- Apply strict, role-based access controls so that everyone involved—alert submitters, verification officers, relief coordinators, and system administrators—clearly understands their responsibilities, and unauthorized actions are prevented.
- Monitor every stage of resource allocation using a relational database, making every decision transparent, traceable, and auditable.

- Maintain data integrity and reliability by ensuring all operations comply with ACID principles, preventing loss or confusion of records.
- Accelerate disaster response and enhance its reliability by eliminating the confusion caused by duplicate, exaggerated, or false alerts.

System architecture

ResQ is a disaster management and resource allocation platform built around clear roles and a layered structure. The main idea is simple that is to take in emergency alerts, check them carefully, and make sure resources get to the right place. What sets it apart or unique is how it tackles bad data—false alarms, scams, or just sloppy information. It does this by mixing hard rules (deterministic checks) with smart AI analysis, so you don't just get a computer guess, but rather a result you can actually trust.

Now separating the architecture into small modules as follows:

1. Identity Management and Role-Based Access Control

ResQ keeps things tight and secure in terms of access of users and what they can do next. Every user gets one role when they join, and the system locks down their permissions.

Here's the list of users:

- Alert Submitter: Sends in disaster info for review.
- Verification Officer: Checks new alerts, looks into the AI analysis, and records results.
- Relief Coordinator: Looks at verified alerts and suggests how to hand out help and allocate resources.
- System Administrator: Sets up top-level accounts, approves resource moves, and runs audits.

Only System Administrators can create or promote privileged users. The system checks every action against the user's permissions—no shortcuts, no exceptions.

2. Disaster Alert Ingestion and Feature Engineering

When someone submits a disaster alert, the system does not add it to database. Instead, it runs a small routine before letting the AI touch anything. The Feature Engineering Module digs into the alert and pulls out key details, like:

- Source Reliability: How trustworthy is the sender, based on past records (if any)?
- Spatial-Temporal Correlation: Does the alert match up with others from the same area and time?
- Deterministic Risk Indicators: Does the message use words or patterns that usually signal real danger?

These features give the AI something solid to work with, making the whole process less vulnerable to tricks or errors.

3. AI-Assisted Verification and Scoring

The AI in the ResQ System doesn't make decisions by itself. Instead, it helps humans decide smarter and faster. Here's how it works: The system bundles the original alert with all those engineered features, then sends it to an external AI service. The AI checks for consistency, plausibility, and urgency. It spits out structured results, sticking to a set format so you never get a vague or confusing answer.

The AI outputs three things:

- A Credibility Score—how likely the alert is real.
- A Priority Score—how urgent and serious the situation is.
- Extracted Entities—the key locations or assets mentioned in the alert.

But as a key note, the AI only provide insights. It never gets to accept or reject an alert on its own and it solely rests in the hand of the verification officer only.

4. Human-in-the-Loop Verification Interface

The ResQ System puts people in charge when it matters most. To stop automated mistakes or random AI guesses from slipping through, every alert goes through a human review. Verification officers see both the original alert and what the AI thinks, all in one place. Nothing gets approved or rejected until a real person confirms it. That confirmation isn't just a click—it's tied directly to the officer's identity and locked in with a timestamp in the system audit trail, so there's never any doubt about who made the call or when. Every decision gets stored in an audit log that can't be changed. If you ever need to look back and see what happened, it's all there.

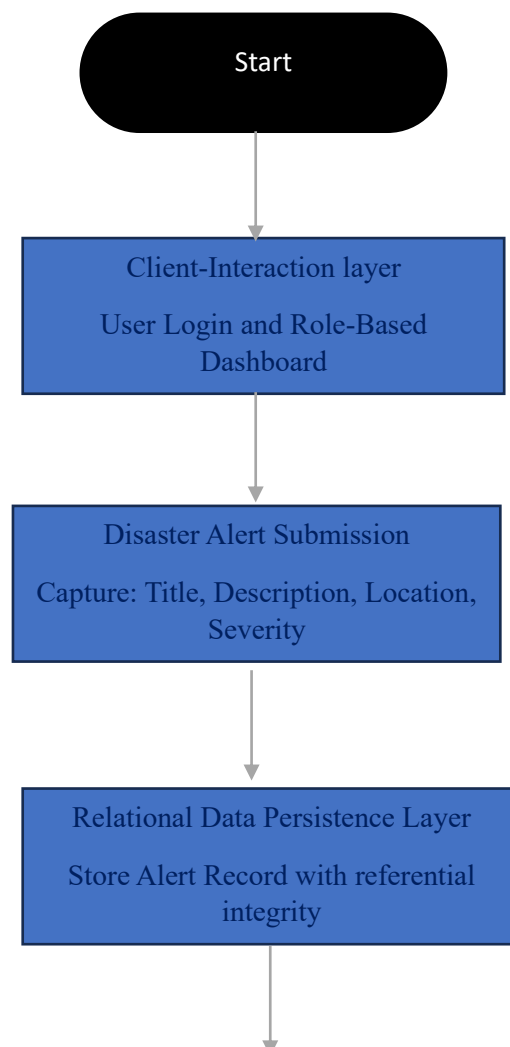
5. Dual-Authorization Resource Allocation

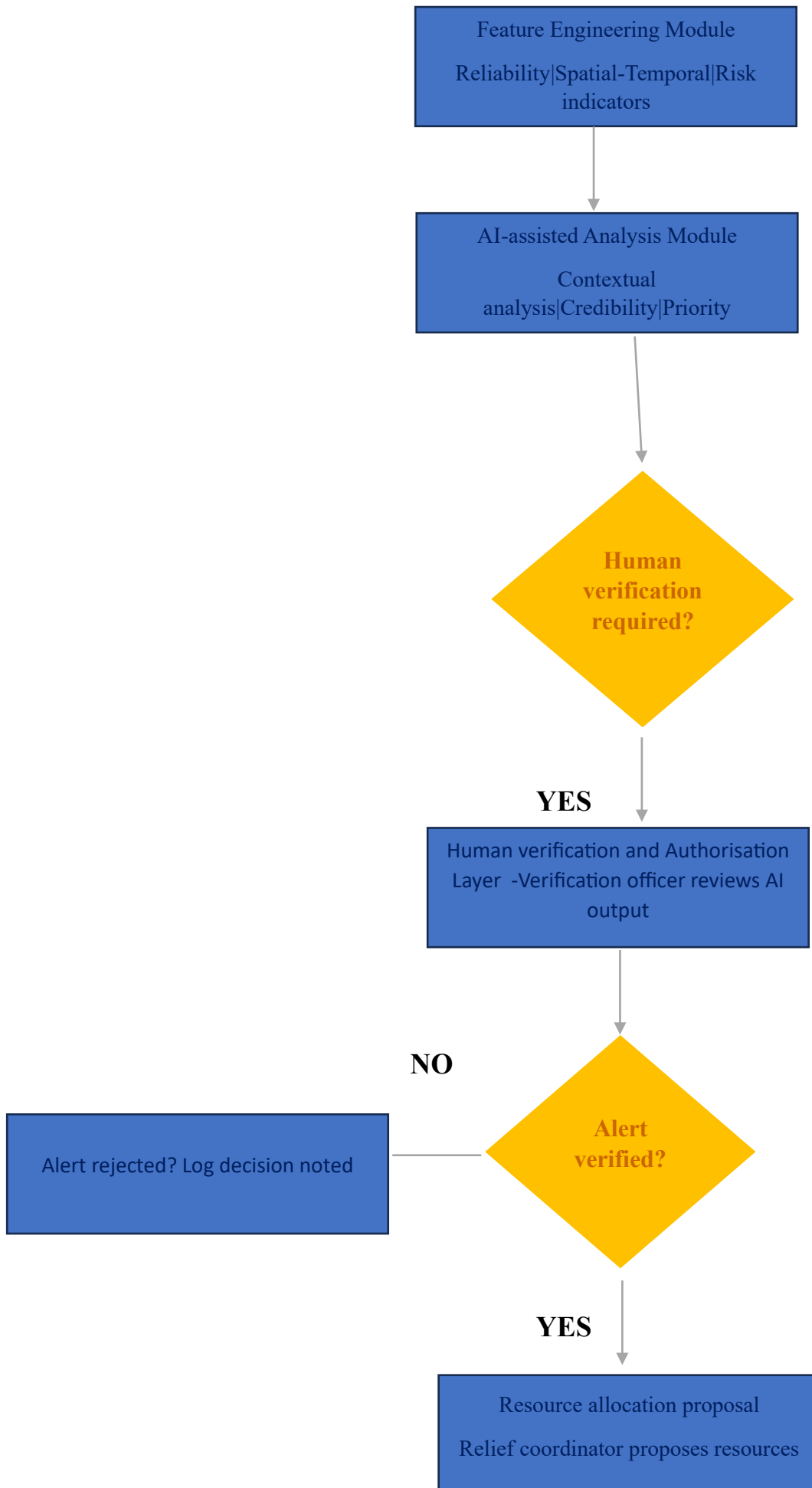
Handing out resources is really serious, so the system makes sure no one person can do it alone. Here's how it works: A relief coordinator suggests where resources or money should go for a verified alert. The system saves this suggestion, along with a required explanation, but nothing moves yet. Then, a system administrator steps in and reviews the proposal. Only after the admin signs off can anything actually get sent out. Every step along the way gets logged for good, so you always know who made each decision.

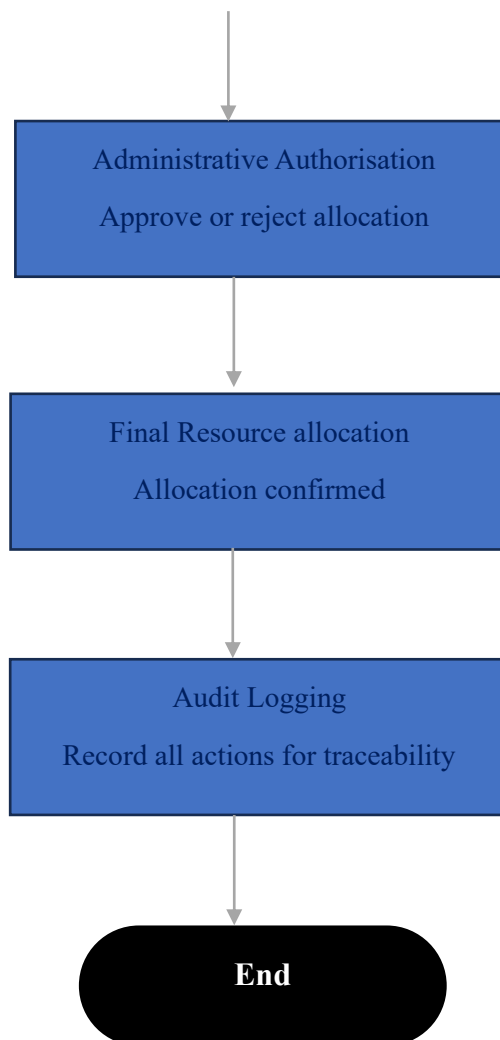
6. Immutable Audit Lodger and Data Integrity Layer

The ResQ System keeps a permanent, tamper-proof record of everything important—alerts, decisions, resource proposals, approvals, and every admin action. Its database links alerts to all their related verification and allocation records, so you never lose the story behind any decision. Admins can pull detailed reports and recreate what happened, step by step. All the data stays safe and consistent, session after session, so you can always trust the integrity of the system and its records.

User workflow diagram:







Summary

This idea introduces an AI-assisted system for verifying disaster alerts and allocating resources, specifically designed to address misinformation, delays, and inadequate accountability in disaster management.

Here's how the system functions in a nutshell: Authorized personnels submit alerts through a secure interface, and these alerts are stored in a relational database. Artificial intelligence analyses each alert for consistency, relevance, and potential warning signs. While the AI provides its assessment, it does not make the final decision—humans retain ultimate authority.

Once alert submitter submits his alert, verification officers review both the alert and the AI's analysis before deciding to approve or reject it. Every action is logged so as to ensure transparency. Once an alert is verified, relief coordinators propose how funds or supplies should be distributed for the specific incident. A system administrator gives the final approval for resource allocation, introducing an added layer of oversight to ensure resources are not disbursed without proper authorization. The entire process is governed by strict role-based controls maintained by a relational database that preserves data integrity and consistency.

By combining AI analysis, human oversight, and a robust data infrastructure, this system effectively limits the chaos caused by false alerts, accelerates authentic responses, and ensures that relief resources are allocated transparently and responsibly.